

ITF23019: Parallel and Distributed Programming

Report for Lab 11: Socket Programming

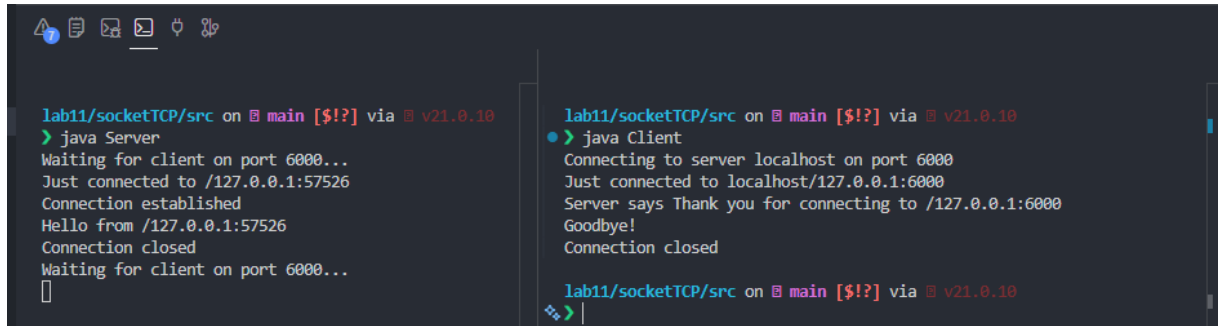
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Exercise 1: SocketTCP

Output:

- **Subtask 1: Run code, include output:**



```

lab11/socketTCP/src on main [$!?] via v21.0.10
> java Server
Waiting for client on port 6000...
Just connected to /127.0.0.1:57526
Connection established
Hello from /127.0.0.1:57526
Connection closed
Waiting for client on port 6000...
[]

lab11/socketTCP/src on main [$!?] via v21.0.10
> java Client
Connecting to server localhost on port 6000
Just connected to localhost/127.0.0.1:6000
Server says Thank you for connecting to /127.0.0.1:6000
Goodbye!
Connection closed

lab11/socketTCP/src on main [$!?] via v21.0.10
>

```

- **Subtask 2: Which class should run first?**
 - o The Server should always run first. The reason for this is because of Socket. Client immediately tries to establish a connection when it runs. This means, if the Server is not running, the connection will fail.
- **Subtask 3: Which port number is used for Client, and which for Server?**
 - o Server: Listens on port: 6000
 - o Client: Assigned port: 67526
- **Subtask 4: What is the IP address or hostname of Client and Server?**
 - o Both utilize localhost but connect via different ports:
 - Server is at 127.0.0.1:6000
 - Client is at 127.0.0.1:57526

Server:

- **Subtask 5: Which line the Server opens the socket?**
 - o The server opens the socket on this line in the constructor (line 8):
serverSocket = new ServerSocket(port);
- **Subtask 6: Which line the Server waits for Client?**
 - o The server waits for a client on this line (line 17):
Socket server = serverSocket.accept();
- **Subtask 7: Which lines the Server creates I/O Streams for communication with the Client?**
 - o The server creates I/O streams on these two lines (lines 21 & 24):
DataInputStream in =
new DataInputStream(server.getInputStream());
DataOutputStream out =
new DataOutputStream(server.getOutputStream());
- **Subtask 8: Which lines the Server performs communication with the Client?**
 - o The server performs communication on these two lines (lines 23 & 26):

```
System.out.println(in.readUTF());
out.writeUTF("Thank you for connecting to " +
    server.getLocalSocketAddress() + "\nGoodbye!");
```

- **Subtask 9: Which line the connection is closed?**

- The connection is closed at line 27 with the following command:

```
server.close();
```

Client:

- **Subtask 10: Which line the Client creates a socket object?**

- The Client creates a socket object at line 11:

```
Socket client = new Socket(serverName, port);
```

- **Subtask 11: Which lines the Client creates I/O Streams for communication with the Client?**

- Client creates I/O Streams at lines 14 and 15 (out), and 18 and 19 (in):

```
OutputStream outToServer = client.getOutputStream();
```

```
DataOutputStream out = new DataOutputStream(outToServer);
```

```
InputStream inFromServer = client.getInputStream();
```

```
DataInputStream in = new DataInputStream(inFromServer);
```

- **Subtask 12: Which lines the Client performs communication with the Client?**

- Client performs communication at line 17 and 21:

```
out.writeUTF("Hello from " + client.getLocalSocketAddress());
```

```
System.out.println("Server says " + in.readUTF());
```

- **Subtask 13: Which line the connection is closed?**

- The Client closed the connection at line 22:

```
client.close();
```

Finally:

- **Subtask 14: How do you relate the above operations of Client and Server to the Three-way handshake data transmission in TCP protocol?**

- First the client tries to connect to the server, then the server accepts the connection. Lastly, they exchange data through input and output streams.

Exercise 2: SocketUDP

Subtask 1: Run the code and include output:

```
lab11/socketUDP/src on 1 main [!?] via v25.0.2 took 55s
> javac Server.java & java Server
[1] 24798
Waiting for client on port 9000
[1] + done      javac Server.java
Received from client: Hello worl
Message sent from IP address: /127.0.0.1 and port: 56224

lab11/socketUDP/src on 1 main [!?] via v25.0.2
> javac Client.java & java Client
[1] 24861
Received from server: Welcome from server
lab11/socketUDP/src on 1 main [!?] via v25.0.2
+ >
```

Subtask 2: Which class should run first?

- This is the same situation as in Exercise 1. The Server should always run first. The reason for this is because of Socket. Client immediately tries to establish a connection when it runs. This means, if the Server is not running, the connection will fail.

Subtask 3: Create the output similar to the figure below. How and why can the server receive multiple requests with multiple ports from client(s)?

- To recreate the output to the provided figure, I had to do the following
 1. Change server-port to 9802 from 9000
 2. Change receivingData byte size from 10 to 11 (Hello world was cut)
- The server can receive multiple requests with multiple port from client(s) because:
 - o UDP is connectionless with one server socket which can handle unlimited clients. Each client is independent.
 - o Each client gets a port from the OS when Socket is created, so there is no port conflict.

```
lab11/socketUDP/src on 1 main [!?] via v21.0.10
> javac Server.java && java Server
Waiting for client on port 9802
Received from client: Hello world
Message sent from IP address: /127.0.0.1 and port: 49139
Received from client: Hello world
Message sent from IP address: /127.0.0.1 and port: 35376
Received from client: Hello world
Message sent from IP address: /127.0.0.1 and port: 53580
Received from client: Hello world
Message sent from IP address: /127.0.0.1 and port: 49761
█
```

Exercise 3: Multicast**Subtask 1:** Reproduce Figures 6 and 7

- Figure 6: Code changes are pushed

```
lab11/ipMulticast/src on main [?!?] via v21.0.10
● > javac MulticastSender.java && java MulticastSender
Message sent to multicast group: 224.0.0.3

lab11/ipMulticast/src on main [?!?] via v21.0.10
○ > 
```

- Figure 7: Code changes are pushed

```
lab11/ipMulticast/src on main [?!?] via v21.0.10
● > javac MulticastReceiver.java && java MulticastReceiver
Note: MulticastReceiver.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

Waiting for client on address: 224.0.0.3
Received from client: Hello, This is a multicast message!
Message sent from IP address: /172.18.98.94 and port: 53683

lab11/ipMulticast/src on main [?!?] via v21.0.10 took 3s
○ > 
```

Subtask 2: Can we set up multiple senders in the same way with setting multiple receivers? Explain why and why not.

- Yes. Multicast supports many senders and many receivers on the same group address and port, but not always.
 - Why:
 - Senders send to one multicast group
 - Any receivers that joined the group gets the packets
 - Why not always:
 - Network/router may block multicast
 - UDP gives no guaranteed delivery/order
 - Same-host multiple receivers may need socket reuse.